databricks

Stop the quiet lapse

Predicting loyalty lapse 60 days earlier
Specialty apparel retail | November 2026

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The problem in one line

Our best members do not cancel. They just stop coming back, and we find out 90 days later.

- 400K loyalty members. Gold and Silver are 30% of members and 67% of trailing-12-month spend.
- $27.1M of trailing-12-month spend now sits with members who have not bought in 90 days or more; $13.9M of it is Gold and Silver.
- The second half of 2026 brought a lapse wave in Gold and Silver members whose home store is in a mall: 38% lapsed, against 27% for the same tiers elsewhere.
- Today's win-back campaign fires at the 90-day mark, when the member is already gone and the offer has to be deep.

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What the data said when we finally looked

Two operational decisions in June and July quietly broke the program for the most valuable segment.

Date	Event	Signal in the data
Jun 1	Core Denim fit change	Denim return rate 7 to 8% in spring, 22% from July on, 70% coded "fit"
Jul 15	Birthday reward cut from $20 to $10	Response rate fell from 34.0% to 13.9%
Jun to Sep	Lapse wave	Mall-store Gold/Silver active buyers down 56% (6,683 to 2,962). Everyone else flat.

The leading indicator was there all along: members go quiet in the app and email 30 to 60 days before their last purchase.

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What we built

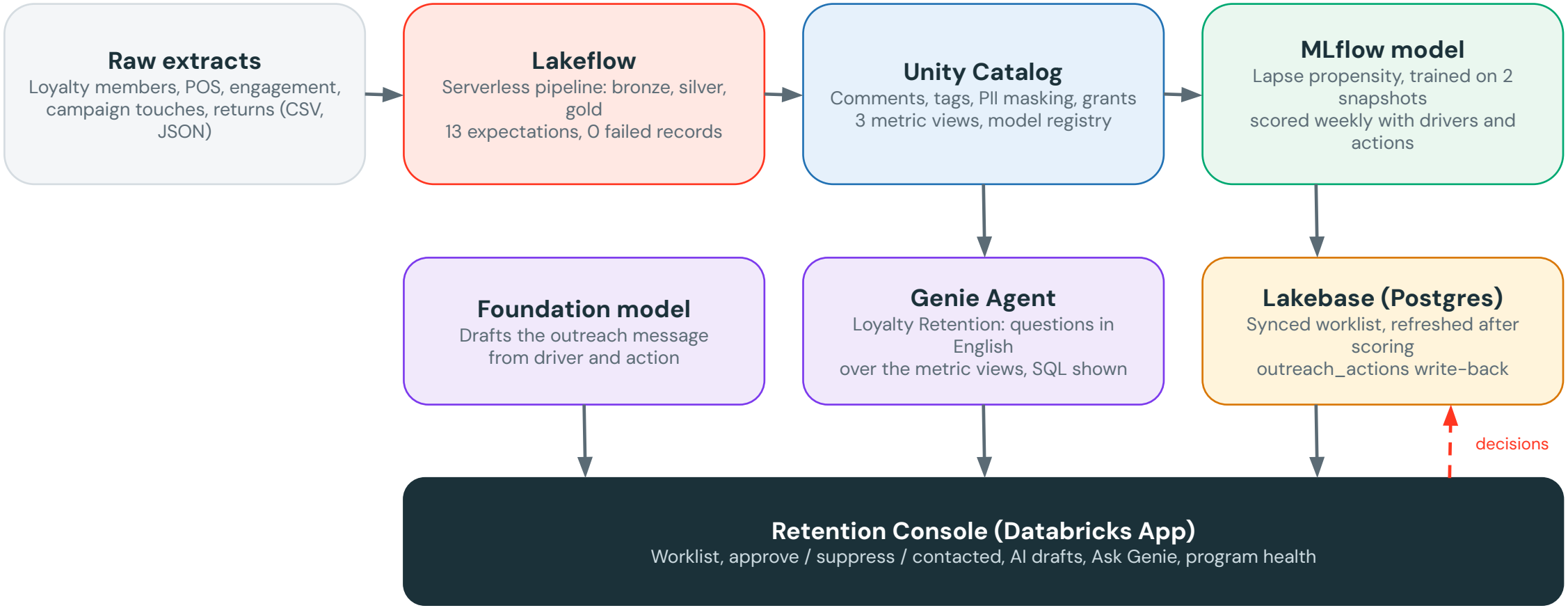
An integrated data journey, not another dashboard. Each stage feeds the next and the retention team closes the loop.

- Lakeflow: five raw extracts into bronze, silver and gold; 13 quality rules, 0 failed records.
- Unity Catalog: every gold column described, PII masked, certified tables, metric views that define lapse rate and revenue at risk once.
- MLflow model: probability a regular shopper buys nothing in the next 90 days, with drivers and a recommended action.
- Lakebase: the ranked worklist served from Postgres; every approve, suppress or contacted decision written back.
- Genie Agent: the CRM team's questions answered in English against governed metrics, SQL shown.
- Retention Console (Databricks App): where the team works the list; a foundation model drafts the outreach.

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How it fits together

Every arrow is a real dependency in the bundle; the console reads three systems and writes decisions back.



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The model, in business terms

Trained on two historical snapshots (Apr 30 and Jun 30), judged on a third it had never seen (Jul 31, outcomes Aug to Oct).

Metric	Result	What it means
Hold-out AUC	0.72	Ranks members well above chance on data it never trained on
Lift in top decile	1.9x	The top 10% of the list lapses at 59%, against a 31% base rate
Lapsed spend captured	$2.4M, 15.6% of all lapsed spend	Found in 10% of the outreach volume
Regular shoppers scored	101,089	3 or more purchases in the last 180 days
High risk	1,408 members, $1.18M at risk	89% are Gold or Silver at mall stores: the model found the cohort on its own
High + Medium risk	37,703 members, $6.15M at risk	The addressable list for a quarter of retention work

Calibration is honest: expected calibration error of 2.3 points, and members scored above 0.6 lapsed at 89% or more. Top drivers: days since last engagement, 12-month purchase count, tier, home store format, lifetime purchases, denim share.

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What the retention team sees

The console turns a score into a decision.

Work the list

- Worklist from Lakebase: member, tier, p(lapse), driver, recommended action, value at risk, sorted by dollars. Milliseconds per query.
- Specific actions: fit-guarantee exchange, restored $20 birthday reward, app push for the disengaging, channel switch for the unresponsive.
- Write-back: approve, suppress or contacted. Every decision lands in Lakebase with who, when and why.

Ask, draft, monitor

- Ask Genie: "Which Gold members in mall stores are at high risk and what should we do?" answered from governed metrics, SQL shown.
- Draft outreach with AI: push, SMS, email or associate talking points grounded in the driver and action. Never mentions risk or scores.
- Program health dashboard for leadership: same KPIs, both 2026 events marked, revenue at risk by driver.

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Value for the executive sponsor

Framed in the KPIs the loyalty P&L is measured on.

KPI	Baseline	With early intervention	Basis
Lapsed-member spend (trailing 12M)	$27.1M	Target 15% recovered: $4.1M a year	Win-back rates of 10 to 20% are typical before the lapse, not after
Incentive cost per saved member	Deep discount after 90 days	Lighter, targeted offer at day 30 to 60	Earlier contact needs a smaller incentive
Campaign response rate	13.9% after the reward cut	Back toward 34% for the devalued segment	Measured before and after in our own data
Time to detect a segment problem	About 90 days	Days	Weekly scoring plus governed metric views

Decision requested: fund a 90-day pilot with the CRM team on the High-risk Gold/Silver mall segment (1,260 members, $1.07M at risk), with a holdout group so the lift is measured, not assumed.

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Value for the domain owner

For the Director of CRM this changes the Monday morning.

- A ranked list with a reason and an action, not a 400K-row export.
- Sized to the team: the High band alone is 1,408 members; filtered to Gold/Silver mall it is 1,260. Add Medium when capacity allows.
- Suppress and contacted flags stop double-touching and carry into next week's list.
- Genie replaces the ad-hoc SQL queue: lapse rate by tier, revenue at risk by driver, response before and after the reward cut, in seconds with the SQL visible.
- Governance is already done. The same "lapse rate" appears in the console, in Genie and in finance's reporting.

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Why Databricks

One platform carried the whole journey, which is why it took hours rather than quarters.

- Serverless Lakeflow pipeline with declarative quality rules and no cluster management.
- Unity Catalog metric views: define each KPI once and every consumer (Genie, app, dashboard) gets the same number.
- MLflow and the UC model registry: versioned, aliased, auditable model promotion.
- Lakebase: Postgres for the operational worklist and write-back, synced from Delta with no separate ETL.
- Genie Agent: natural language over governed data, with instructions and benchmark questions checked in like code (4 of 4 correct).
- Databricks Apps: the business surface sits next to the data and inherits its permissions.

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Next steps

A pilot that proves the lift, then widen the loop.

- Pilot: 90 days, High-risk Gold/Silver mall segment, holdout group, measured lift.
- Give merchandising the fit-return signal weekly so the next fit problem is caught in days.
- Scoring already runs weekly with the Lakebase worklist refreshing automatically; add drift monitoring.
- Extend the Genie Agent to store operations and merchandising questions on the same governed layer.
- Code, run logs and committed outputs: github.com/OrrinDatabricks/loyalty-lapse-propensity



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